

Final Presentation

Advanced Software Engineering

s1300137 Anri Sakamaki

s1310257 Hiroto Ohori

GOURP:TETRIS

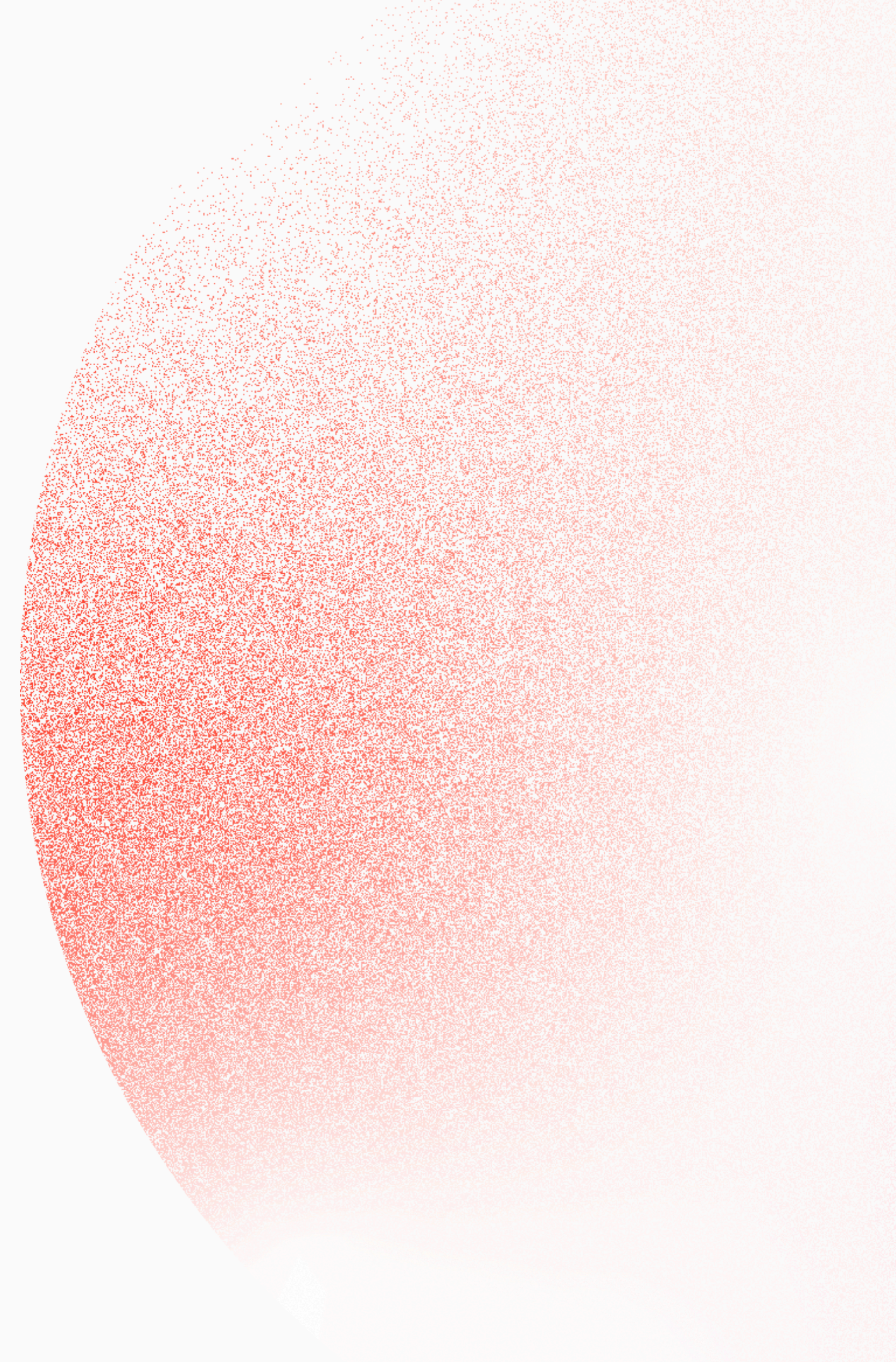


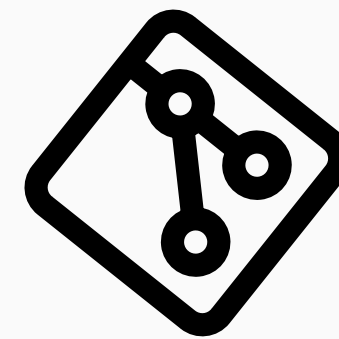
Table of Contents

Requirments and Tech Task	3
System Roles	4
Processing Flow & Backend– Frontend Communication	5
Design Philisophy	6
Demonstration	7

Requirements and Tech tasks

Requirements

- **Frontend:** React
- **Backend:** C
- **Communication:** HTTP with JSON format
- **Version control:** Git and Github



Tech task

- **Several sorting algorithms** are implemented
- **A GUI** is required to help users learn sorting algorithms easily
- **Software validity** is ensured by data generators and testing engine

System Roles

Backend

- Executes sorting algorithms
- Records each operation (comparison, swap, assignment) as logs
- Returns the sorting process in JSON format

Frontend

- Provides the user interface
- Sends sorting requests to the backend
- Visualizes the sorting process step by step

Processing Flow & Backend–Frontend Communication

Processing Flow

- 1. The user selects a sorting algorithm and inputs an array**
- 2. The frontend sends a request to the backend**
- 3. The backend performs the sorting and generates logs**
- 4. The frontend receives the logs and visualizes the process**

Backend–Frontend Communication

- Communication is done via HTTP POST**
- Data is exchanged in JSON format**
- The backend returns the initial array and sorting steps as logs**

Design philosophy

Extensibilit

Add sorting algorithm in sort file
in order to put new sorting system

CHANGING REQUIRED

Frontend: SortList.jsx

Backend: Add a code in sort file

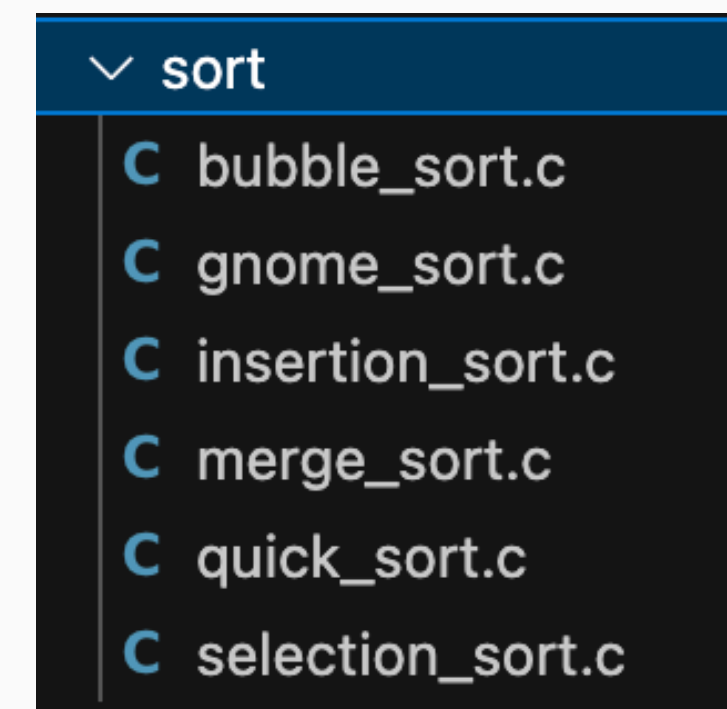
Testengine: SortTestEngine.jsx

SortList.jsx

```
const SORT_METHODS = [  
  { id: "quick", label: "Quick" },  
  { id: "selection", label: "Selection" },  
  { id: "gnome", label: "Gnome" },  
  { id: "bubble", label: "Bubble" },  
];
```

SortTestEngine.jsx

```
const SORT_METHODS = [  
  { id: 'bubble', name: 'Bubble Sort' },  
  { id: 'quick', name: 'Quick Sort' },  
  { id: 'selection', name: 'Selection Sort' },  
  { id: 'gnome', name: 'Gnome Sort' }  
];
```



Sort file
in Backend file

Test engine

Tech Skill

Library

React Router Dom

How to

1. Change router
localhost/test
2. Chose sortings
you want to test
3. Click “Run All Tests”

Sort Test Engine

Comprehensive testing suite for sorting algorithms

Select Algorithms to Test

☒ Bubble Sort

☒ Quick Sort

☒ Selection Sort

☒ Gnome Sort

Run All Tests

Bubble Sort

Pass Rate	100.0%
Passed	✓ 8
Failed	✗ 0
Avg Time	14ms
Total Steps	12,248

Quick Sort

Pass Rate	100.0%
Passed	✓ 8
Failed	✗ 0
Avg Time	13ms
Total Steps	1,914

Selection Sort

Pass Rate	100.0%
Passed	✓ 8
Failed	✗ 0
Avg Time	14ms
Total Steps	6,538

Gnome Sort

Pass Rate	100.0%
Passed	✓ 8
Failed	✗ 0
Avg Time	16ms
Total Steps	9,543

Bubble Sort - Detailed Results

Empty Array 0 elements	✓ PASS	0ms Time	0 Steps
---------------------------	-----------	-------------	------------

7

Demonstration



Demonstration